

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

04

EPSILON-NFA (ϵ -NFA)

DENNIS GUNAWAN

REVIEW

Nondeterministic Finite Automata (NFA):

Nondeterministic Finite Automata (NFA)

Conversion of NFA to DFA

OUTLINE

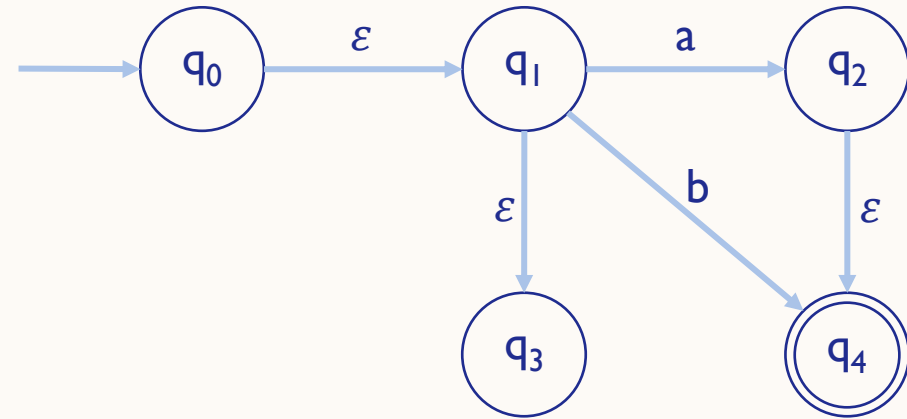
Epsilon-NFA (ϵ -NFA)

Epsilon-Closures

Conversion of ϵ -NFA to NFA

Union and Concatenation of FSA

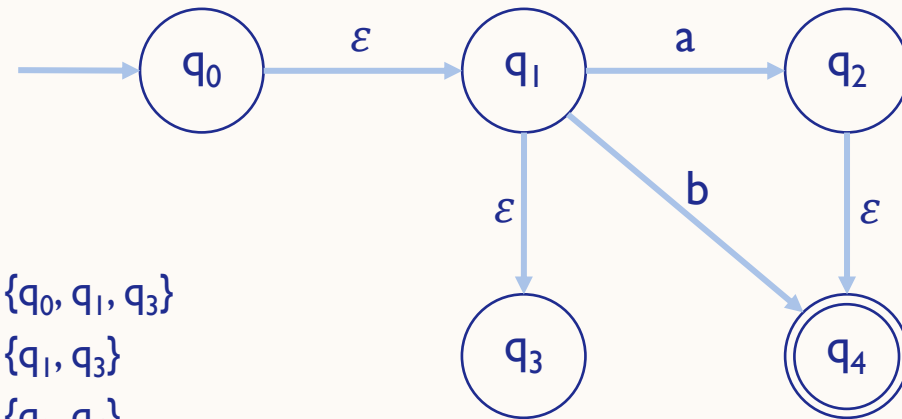
EPSILON-NFA (ϵ -NFA)



- ϵ : the empty string
 - An NFA is allowed to make a transition spontaneously, without receiving an input symbol
- δ : a transition function from $Q \times \Sigma \cup \{\epsilon\} \rightarrow 2^Q$
- Every state on ϵ goes to itself

EPSILON-CLOSURES

- Every state that can be reached from a particular state along any path whose arcs are all labeled ϵ



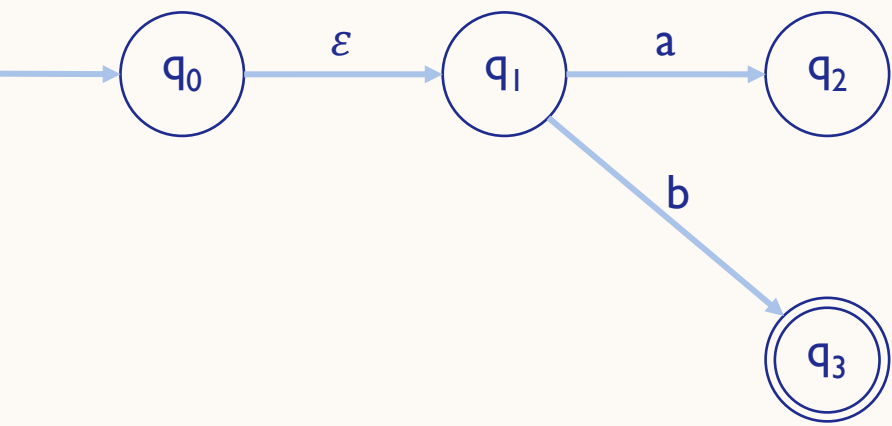
- Example
 - ϵ -closure(q_0) = $\{q_0, q_1, q_3\}$
 - ϵ -closure(q_1) = $\{q_1, q_3\}$
 - ϵ -closure(q_2) = $\{q_2, q_4\}$
 - ϵ -closure(q_3) = $\{q_3\}$
 - ϵ -closure(q_4) = $\{q_4\}$

CONVERSION OF ϵ -NFA TO NFA

- $\delta'(\text{state}, \text{input}) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(\text{state}), \text{input}))$
- $F' = F \cup \{q \mid (\epsilon\text{-closure}(q) \cap F) \neq \emptyset\}$

CONVERSION OF ε-NFA TO NFA

Convert the following ε-NFA to its equivalent NFA



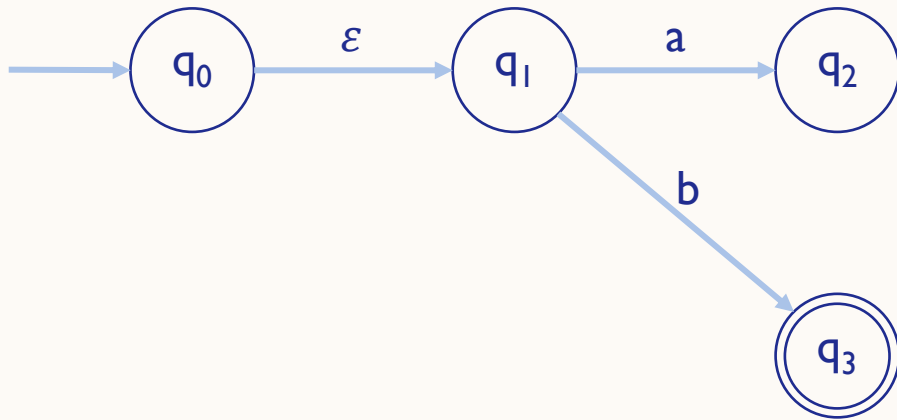
δ'	a	b
q_0	$\{q_2\}$	$\{q_3\}$
q_1	$\{q_2\}$	$\{q_3\}$
q_2	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$

δ'	ϵ -closure	δ , input: a	ϵ -closure(δ)
q_0	$\{q_0, q_1\}$	$\emptyset$ $\{q_2\}$	$\emptyset$ $\{q_2\}$
q_1	$\{q_1\}$	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\emptyset$	$\emptyset$
q_3	$\{q_3\}$	$\emptyset$	$\emptyset$

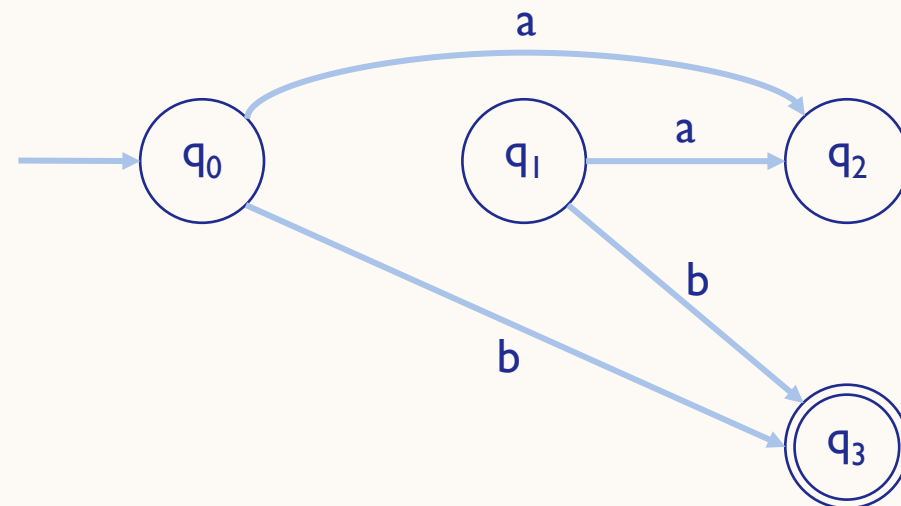
δ'	ϵ -closure	δ , input: b	ϵ -closure(δ)
q_0	$\{q_0, q_1\}$	$\emptyset$ $\{q_3\}$	$\emptyset$ $\{q_3\}$
q_1	$\{q_1\}$	$\{q_3\}$	$\{q_3\}$
q_2	$\{q_2\}$	$\emptyset$	$\emptyset$
q_3	$\{q_3\}$	$\emptyset$	$\emptyset$

CONVERSION OF ϵ -NFA TO NFA

Convert the following ϵ -NFA to its equivalent NFA

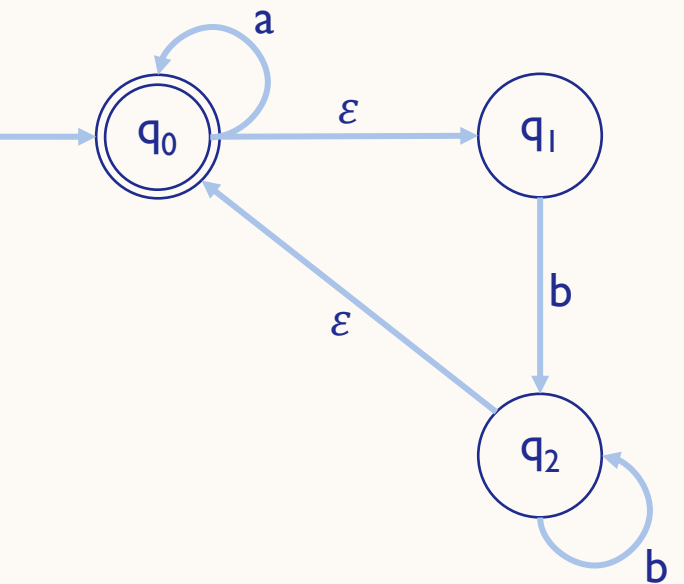


δ'	a	b
q_0	$\{q_2\}$	$\{q_3\}$
q_1	$\{q_2\}$	$\{q_3\}$
q_2	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$



CONVERSION OF ε-NFA TO NFA

Convert the following ε-NFA to its equivalent NFA



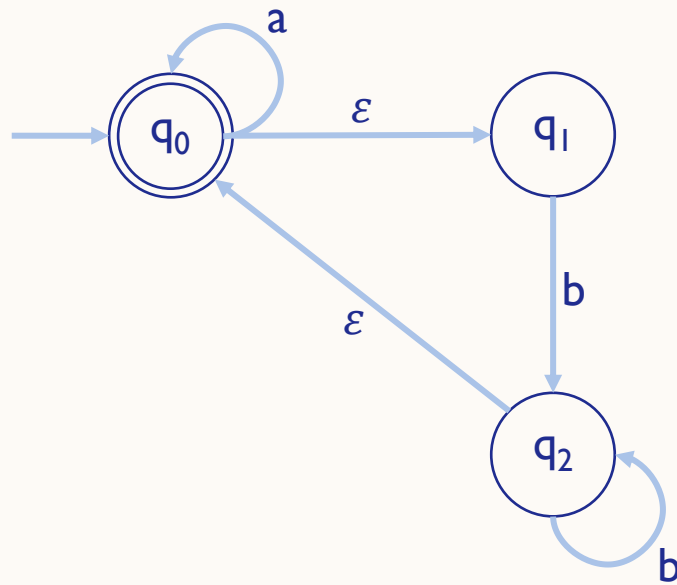
δ'	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$
q_1	$\emptyset$	$\{q_0, q_1, q_2\}$
q_2	$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$

δ'	ϵ -closure	δ , input: a	ϵ -closure(δ)
q_0	$\{q_0, q_1\}$	$\{q_0\}$ $\emptyset$	$\{q_0, q_1\}$ $\emptyset$
q_1	$\{q_1\}$	$\emptyset$	$\emptyset$
q_2	$\{q_0, q_1, q_2\}$	$\{q_0\}$ $\emptyset$ $\emptyset$	$\{q_0, q_1\}$ $\emptyset$ $\emptyset$

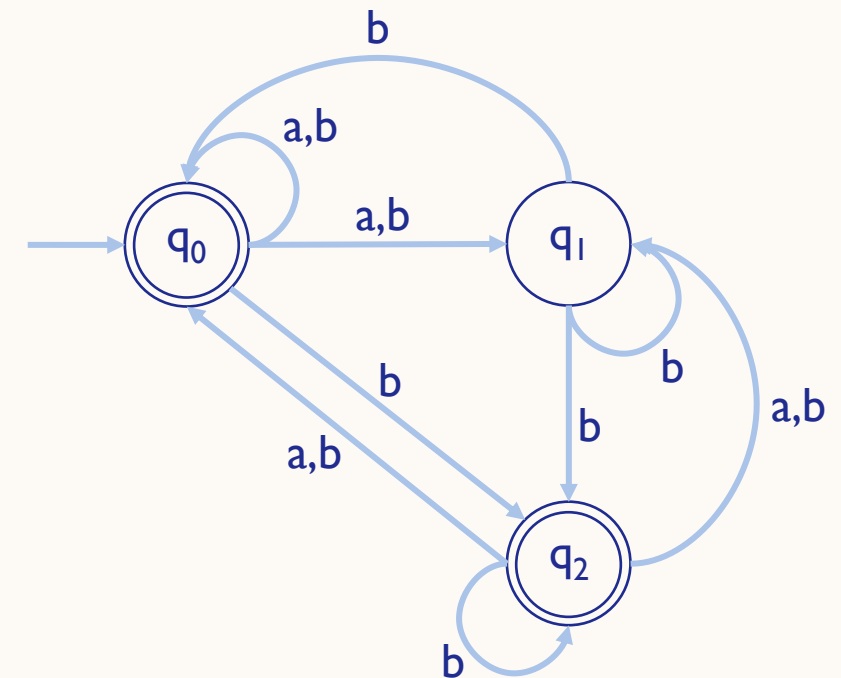
δ'	ϵ -closure	δ , input: b	ϵ -closure(δ)
q_0	$\{q_0, q_1\}$	$\emptyset$ $\{q_2\}$	$\emptyset$ $\{q_0, q_1, q_2\}$
q_1	$\{q_1\}$	$\{q_2\}$	$\{q_0, q_1, q_2\}$
q_2	$\{q_0, q_1, q_2\}$	$\emptyset$ $\{q_2\}$ $\{q_2\}$	$\emptyset$ $\{q_0, q_1, q_2\}$ $\{q_0, q_1, q_2\}$

CONVERSION OF ϵ -NFA TO NFA

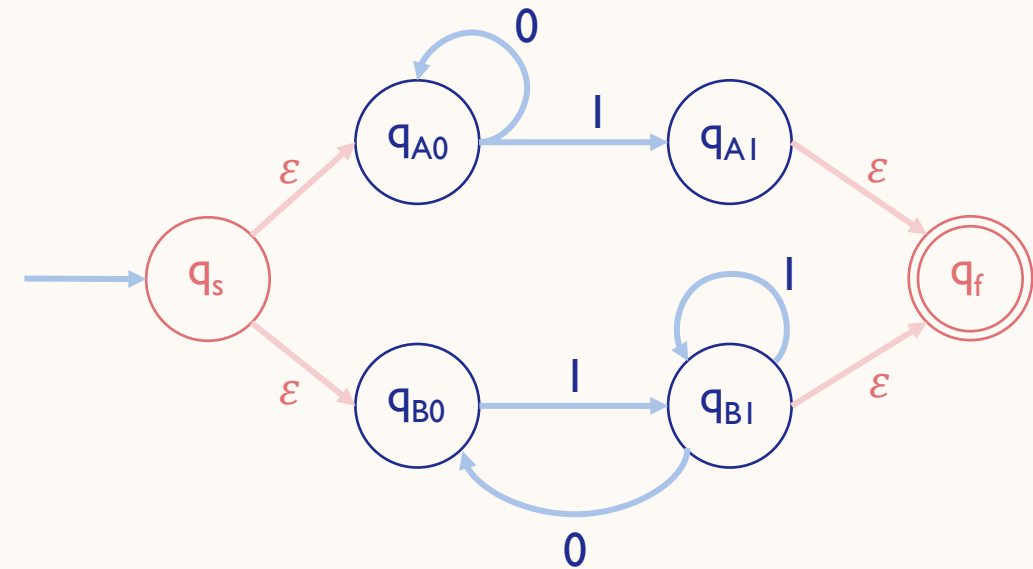
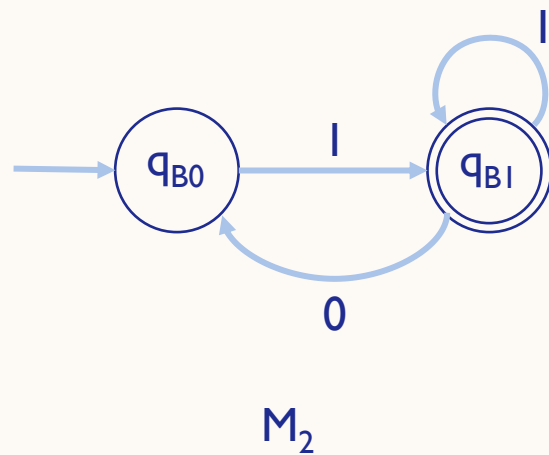
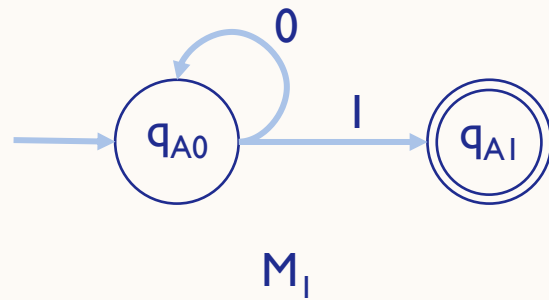
Convert the following ϵ -NFA to its equivalent NFA



δ'	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$
q_1	$\emptyset$	$\{q_0, q_1, q_2\}$
q_2	$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$

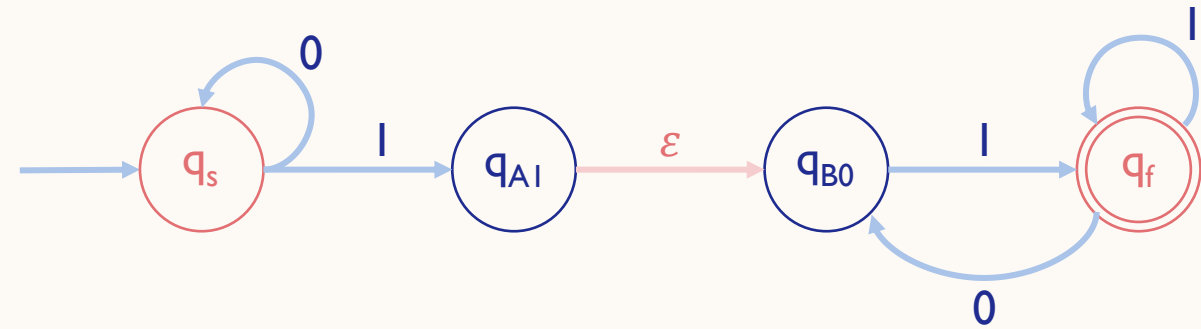
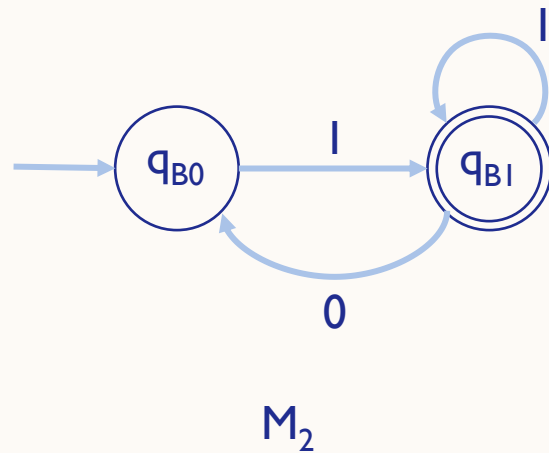
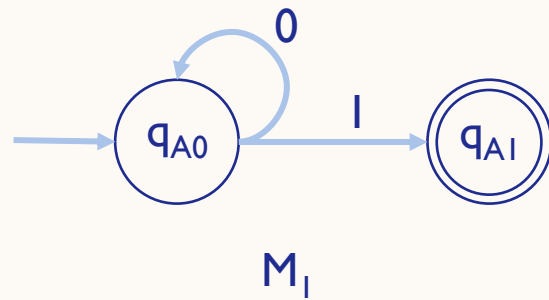


UNION OF FSA



$$M_3; L(M_3) = L(M_1) \cup L(M_2) = L(M_1) + L(M_2)$$

CONCATENATION OF FSA



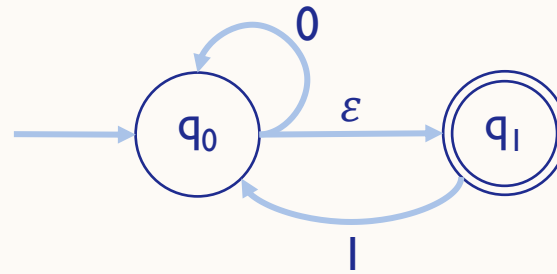
$$M_4; L(M_4) = L(M_1) L(M_2)$$



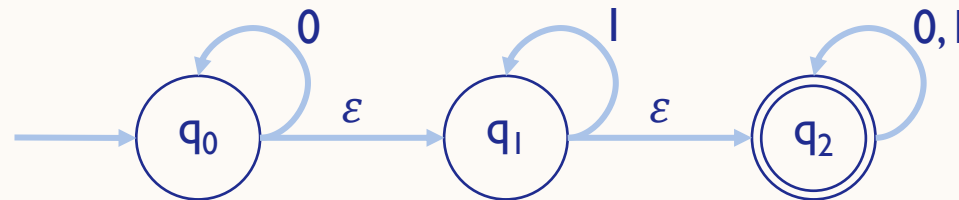
PRACTICE

EXERCISES

1. Convert the following ϵ -NFA to its equivalent NFA.

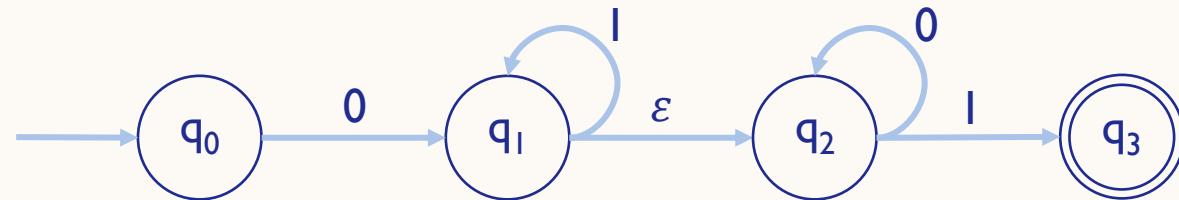


2. Convert the following ϵ -NFA to its equivalent NFA.

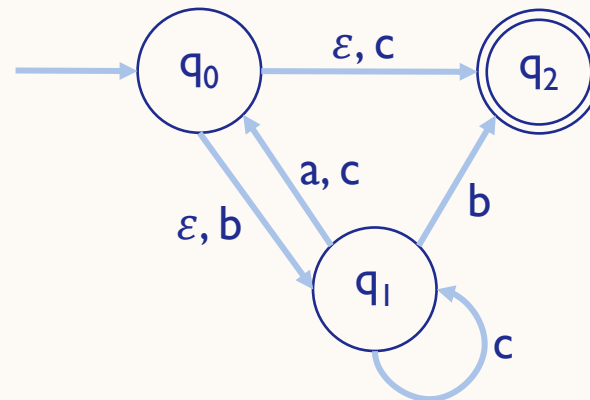


EXERCISES

3. Convert the following ϵ -NFA to its equivalent NFA.

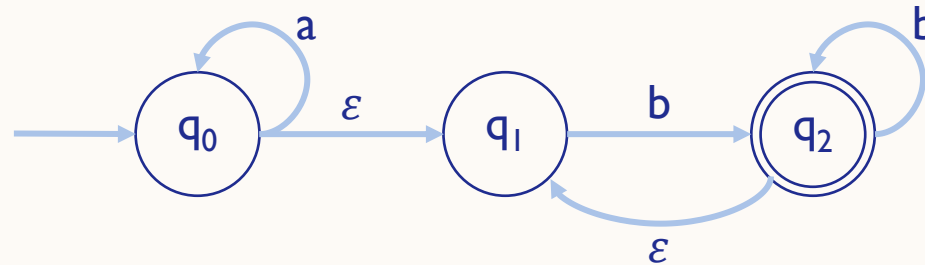


4. Convert the following ϵ -NFA to its equivalent NFA.

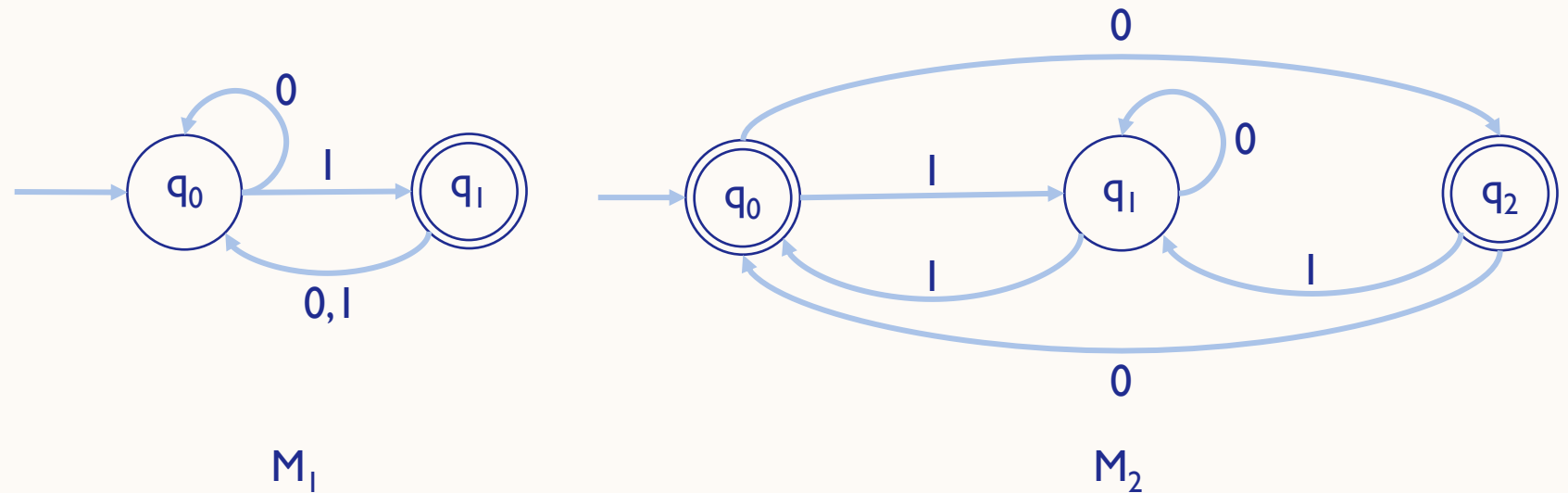


EXERCISES

5. Convert the following ϵ -NFA to its equivalent NFA.



EXERCISES



6. Construct M_3 that accepts the language $L(M_3)$ where $L(M_3) = L(M_1) + L(M_2)$.
7. Construct M_4 that accepts the language $L(M_4)$ where $L(M_4) = L(M_1) L(M_2)$.

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

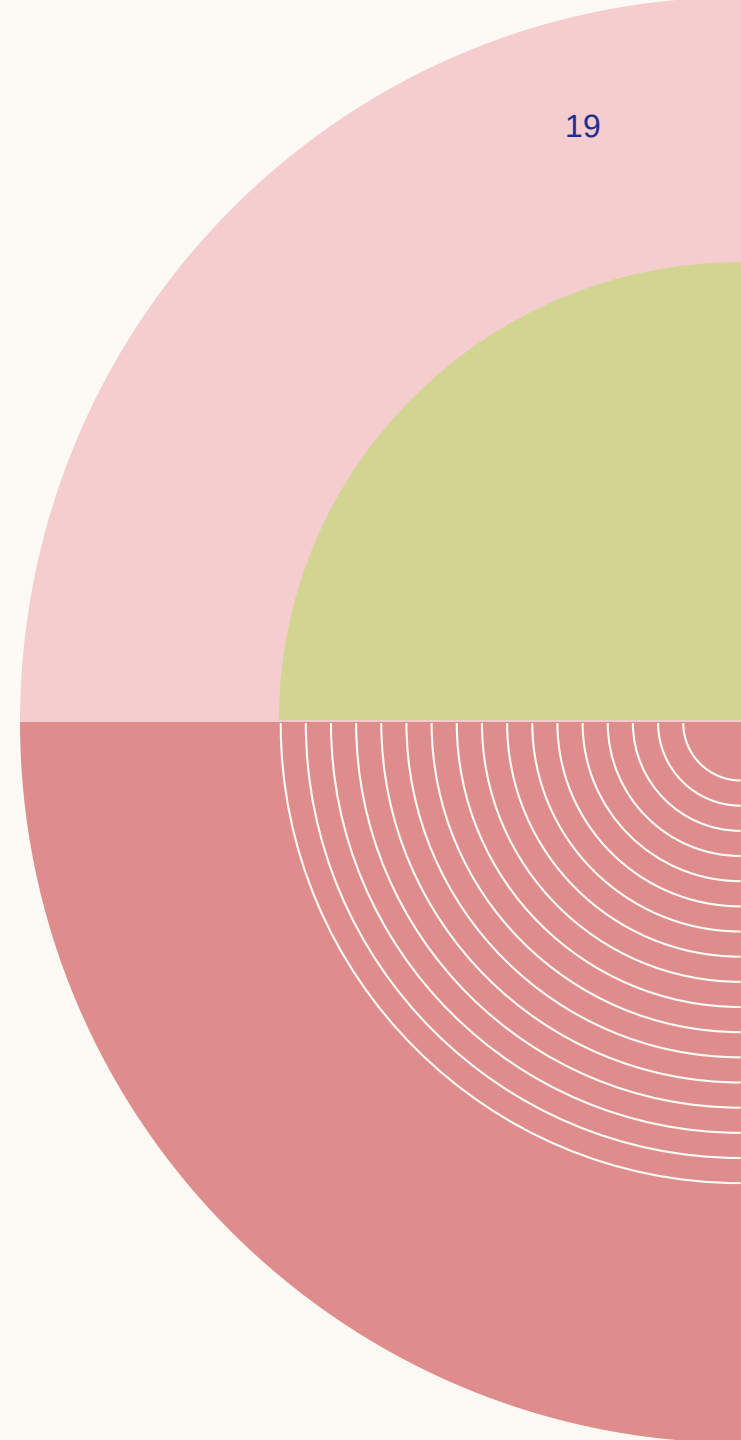
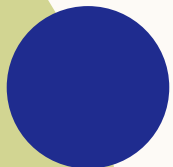
NEXT

Minimization of DFA:

Minimization of DFA

Partitioning Method

Table Filling Method (Myhill Nerode Theorem)



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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